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Problem

LORA UNDER MULTI-TASK LOAD

- LoRA is efficient but still stores and optimizes both low-rank matrices for every task.
- Directly merging heterogeneous adapters causes parameter interference and large performance drops.
- Sequential adaptation can overwrite safety alignment through catastrophic forgetting.

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Core idea

FREEZE A, SPARSIFY B

- Keep A as an independent random projection for each task; learn only how B combines that subspace.
- A calibration run ranks all B entries globally by magnitude and selects a task-specific sparse mask.
- Reset B, then train only the top entries; 90% sparsity leaves just 10% of B active.

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Single-task method

CALIBRATE ONCE, REUSE

- Collect B across every layer/projection, compute one global quantile threshold and form mask M_t .
- The final update is $\Delta_t = A_t (B_t \text{ elementwise } M_t)$; masks are calibrated once per task.
- Frozen A removes gradients/optimizer states and merges into W_0 with no inference latency.

LORI

LoRI: Reducing Cross-Task Interference in Multi-Task Low-Rank Adaptation

COLM 2025

Freeze LoRA's random projection A and sparsely train B with task-specific masks, reducing parameters while separating task updates for merging and continual learning.

PEFTsparsitymulti-task

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Multi-task theory

ORTHOGONALITY + SPARSITY

- $_F \rightarrow 0$ for independent random A_s, A_t as $d_{in} \rightarrow \text{infinity}$
- Concatenated merging directly sums nearly orthogonal adapters and avoids cross-terms from linear merging.
- Sparse task masks isolate sequential updates; expected overlap at 90% sparsity is about 1%.

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Evidence

95% FEWER THAN LORA

- Llama-3-8B: LoRI-S trains 4.4M parameters (0.05%) vs LoRA's 84M (1.03%).
- HumanEval pass@10: LoRI-D 63.2, LoRI-S 59.6, LoRA 50.8; NLU and safety remain competitive.
- Concatenated LoRI-D merging retains 62.2 pass@10 and 86.6% safety refusal, far above merged LoRA on code.

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Meaning & limits

TAKEAWAY

- Random subspaces reduce merging interference; sparsity regularizes adaptation and limits forgetting.
- A two-phase safety-to-task schedule builds lightweight safety-preserving adapters.
- Current sparsity is unstructured; calibration cost, merging-weight sensitivity and multimodal extension remain open.